E-Commerce Sales Data Analysis

and Interactive Power BI Dashboard

Final Data Analytics Project Report
Week 6 Internship Assignment

Field	Details
Student Name	[Your Name]
Assignment	Week 6: Git, GitHub & Final Data Analytics Project
Total Marks	100
Duration	7 Days
Submission Date	October 25, 2026
Report Generated	October 25, 2026 at 13:19
Dataset Type	Synthetic (Superstore-style, 9,994 records)
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Git

■ **IMPORTANT NOTICE:** The dataset used in this project is synthetically generated using Python (NumPy seed=42) to simulate retail sales data. All business findings are for educational purposes only and do not represent real-world business data.

Abstract

This report presents a comprehensive end-to-end data analytics capstone project analyzing 9,994 e-commerce retail sales transactions spanning January 2020 to December 2023. Using Python (Pandas, NumPy, Matplotlib, Seaborn), the project performs systematic data cleaning, exploratory data analysis, and generates 10 professional data visualizations. Key findings include: a total revenue of $33,170,960, a profit of $3,642,648 (10.98% margin), a strong negative correlation between discount rates and profitability (-0.854), and a structurally loss-making Furniture Tables sub-category. Three actionable business recommendations are provided: implementing discount governance, developing a Corporate segment strategy, and repricing the Tables sub-category. A complete Power BI dashboard preparation package is included with setup instructions and DAX measures. The project is documented for GitHub publication with full reproducibility.

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1. Introduction

E-commerce has become a dominant channel for retail sales globally, generating vast amounts of transaction data that, when analyzed properly, can unlock powerful business insights. This project applies the complete data analytics lifecycle to a retail sales dataset — from raw data ingestion through cleaning, analysis, visualization, and business recommendation.

The project uses a synthetic dataset modeled on the Tableau Sample Superstore, a widely used benchmark dataset in data analytics education. The dataset contains 9,994 transaction records across three product categories (Technology, Office Supplies, Furniture), four US geographic regions, three customer segments, and a four-year period (2020-2023).

The analysis is implemented in Python using the data science stack (Pandas, NumPy, Matplotlib, Seaborn), with an interactive Power BI dashboard for business intelligence reporting. The project is version-controlled using Git and documented for GitHub submission.

2. Problem Statement

E-commerce businesses generate large volumes of sales data but often lack the analytical infrastructure to convert this data into actionable business intelligence. Key unanswered questions include:

• Which product categories and sub-categories are truly profitable?

• How do discount strategies affect overall profitability?

• Which geographic regions and customer segments drive the most value?

• What seasonal patterns should guide inventory and staffing decisions?

• Which products generate high revenue but low (or negative) profit?

Without answering these questions, management risks over-discounting profitable products, under-investing in high-growth segments, and failing to address structurally unprofitable product lines.

3. Project Objectives

1. Analyze total sales, profit, and margin performance across the full dataset

2. Identify high-performing and low-performing products, categories, and sub-categories

3. Analyze monthly and yearly sales trends to identify seasonal patterns

4. Examine regional and customer-segment performance

5. Investigate the quantitative relationship between discount rates and profitability

6. Create at least eight polished Python visualizations

7. Prepare an interactive Power BI dashboard with KPI cards and slicers

8. Generate at least five evidence-based business insights with numerical support

9. Provide at least two specific, actionable business recommendations

10. Publish and document the complete project on GitHub

4. Dataset Description

4.1 Dataset Source

Dataset Type: Synthetic | Generator: src/generate_dataset.py | Random Seed: 42 | Records (raw): 9,997 | Records (cleaned): 9,994 | Columns (raw): 21 | Columns (cleaned): 26

The dataset was generated to closely simulate the statistical properties of the Tableau Sample Superstore dataset, a publicly available retail analytics benchmark. The actual Tableau dataset is available at: https://www.tableau.com/sites/default/files/2021-05/Sample%20-%20Superstore.xls

4.2 Dataset Schema

Column	Type	Description
Order ID	String	Unique order identifier (multiple rows per order)
Order Date	Date	Date order was placed (2020-01-01 to 2023-12-31)
Ship Date	Date	Date order was shipped (nullable)
Ship Mode	String	Standard Class / Second Class / First Class / Same Day
Customer ID	String	Unique customer identifier
Segment	String	Consumer / Corporate / Home Office
Region	String	East / West / Central / South
Category	String	Technology / Office Supplies / Furniture
Sub-Category	String	17 sub-categories across 3 categories
Product Name	String	Product display name
Sales	Float	Transaction revenue (USD)
Quantity	Integer	Units sold
Discount	Float	Discount rate (0.0 to 1.0)
Profit	Float	Net profit (USD, can be negative)
Profit Margin %	Float	Profit/Sales × 100 (derived)

5. Tools and Technologies

Tool/Library	Version	Purpose
Python	3.13.15	Primary programming language
Pandas	2.x	Data manipulation and analysis
NumPy	2.x	Numerical computing and array operations
Matplotlib	3.11.x	Data visualization (base library)
Seaborn	0.13.x	Statistical visualizations and heatmaps
python-pptx	1.0.x	PowerPoint presentation generation
reportlab	5.x	PDF report generation
nbformat	5.x	Jupyter notebook creation
Power BI Desktop	2.120+	Interactive dashboard (manual)
Git	2.56.x	Version control
GitHub	—	Remote repository and documentation
Jupyter Notebook	7.x	Interactive EDA environment

6. Data Cleaning Methodology

Data cleaning was performed by `src/data_cleaning.py`. The process followed a nine-step pipeline documented in `reports/data_cleaning_summary.md`.

6.1 Cleaning Results Summary

Metric	Before Cleaning	After Cleaning	Change
Total Rows	9,997	9,994	-3
Total Columns	21	26	+5 (derived)
Missing Values	12 (Ship Date)	12 (retained)	—
Duplicate Rows	3	0	-3
Invalid Sales (≤0)	0	0	—
Date Format	String DD/MM/YYYY	datetime64	Parsed

6.2 Key Cleaning Decisions

Missing Ship Dates: 12 records with NULL Ship Dates were retained. These represent legitimate unshipped orders and should not be removed.

Outlier Treatment: 1,733 potential profit outliers were identified using the ±3×IQR rule. These were retained because large bulk orders and promotional campaigns legitimately produce extreme values in retail datasets. Removing them would introduce bias into aggregated statistics.

Derived Features: Five new columns were added: Order Year, Order Month, Order Month Name, Order Quarter, and Profit Margin %. These enable temporal analysis and margin calculations without modifying the original data.

7. Exploratory Data Analysis

7.1 Key Performance Indicators

KPI	Value
Total Sales	$33,170,959.70
Total Profit	$3,642,648.41
Total Quantity Sold	74,555 units
Total Unique Orders	9,070
Unique Customers	800
Overall Profit Margin	10.98%
Average Order Value	$3,657.22

7.2 Sales by Category

Category	Total Sales	Total Profit	Profit Margin %	Transactions
Technology	$25,018,661.95	$4,199,956.68	16.79%	3,389
Furniture	$6,855,598.45	$-644,293.37	-9.40%	3,297
Office Supplies	$1,296,699.30	$86,985.10	6.71%	3,308

7.3 Regional Performance

Region	Total Sales	Total Profit	Profit Margin %	Transactions
West	$10,054,303.15	$1,217,331.24	12.11%	2,977
East	$9,280,839.25	$958,615.62	10.33%	2,777
South	$7,206,530.14	$884,451.45	12.27%	2,007
Central	$6,629,287.16	$582,250.10	8.78%	2,233

7.4 Customer Segment Performance

Segment	Total Sales	Share %	Transactions
Consumer	$17,591,638.00	53.0%	5,205
Corporate	$10,175,374.43	30.7%	2,995
Home Office	$5,403,947.27	16.3%	1,794

8. Descriptive Statistics

Statistic	Sales	Profit	Quantity	Discount	Margin %
Mean	$3,319.09	$364.48	7.46	0.1385	1.34%
Median	$962.62	$14.74	7.0	0.1000	4.58%
Std Dev	$6,317.44	$2,034.98	4.04	0.1668	23.36%
Min	$3.38	$-7,515.12	1	0.00	-113.32%
Max	$61,198.74	$24,133.49	14	0.80	48.47%
Q25	$267.79	$-75.31	4.0	0.00	-9.63%
Q75	$3,257.07	$134.61	11.0	0.20	17.73%

9. Correlation and Outlier Analysis

9.1 Correlation Matrix

	Sales	Profit	Quantity	Discount	Margin %
Sales	1.000	0.785	0.286	-0.103	0.217
Profit	0.785	1.000	0.104	-0.259	0.431
Quantity	0.286	0.104	1.000	0.009	-0.002
Discount	-0.103	-0.259	0.009	1.000	-0.854
Profit Margin %	0.217	0.431	-0.002	-0.854	1.000

9.2 Key Correlation Findings

Sales ↔ Profit: r = 0.785 — Strong positive. Higher sales reliably predict higher absolute profit. This is expected in a well-priced product portfolio.

Discount ↔ Profit Margin %: r = -0.854 — Strong negative. This is the most important finding in the dataset. As discount rates increase, profit margins collapse proportionally. Transactions at 30-50% discount frequently produce negative margins.

Discount ↔ Profit: r = -0.259 — Moderate negative. The moderate strength (vs. the strong Discount-Margin correlation) reflects that high-sales transactions can absorb more discount before going negative.

Note: Correlation indicates statistical association, not causation. However, in the case of discounts and margins, the mechanism is mathematically direct — discounts reduce revenue while costs remain constant.

9.3 Outlier Analysis

Method: Interquartile Range (IQR), threshold ±3×IQR applied to Profit.

Variable	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers
Sales	$268	$3,257	$2,989	$-8,700	$12,225	690
Profit	$-75	$135	$210	$-705	$764	1,732

Decision: All outliers RETAINED. Extreme sales and profit values in retail datasets represent genuine large-volume or high-value transactions (e.g., bulk Copier purchases for Corporate clients). Removing them would introduce downward bias in aggregate revenue and profit calculations.

10. Data Visualizations

Ten professional visualizations were generated using Matplotlib and Seaborn. All charts use real data values from cleaned_sales.csv.

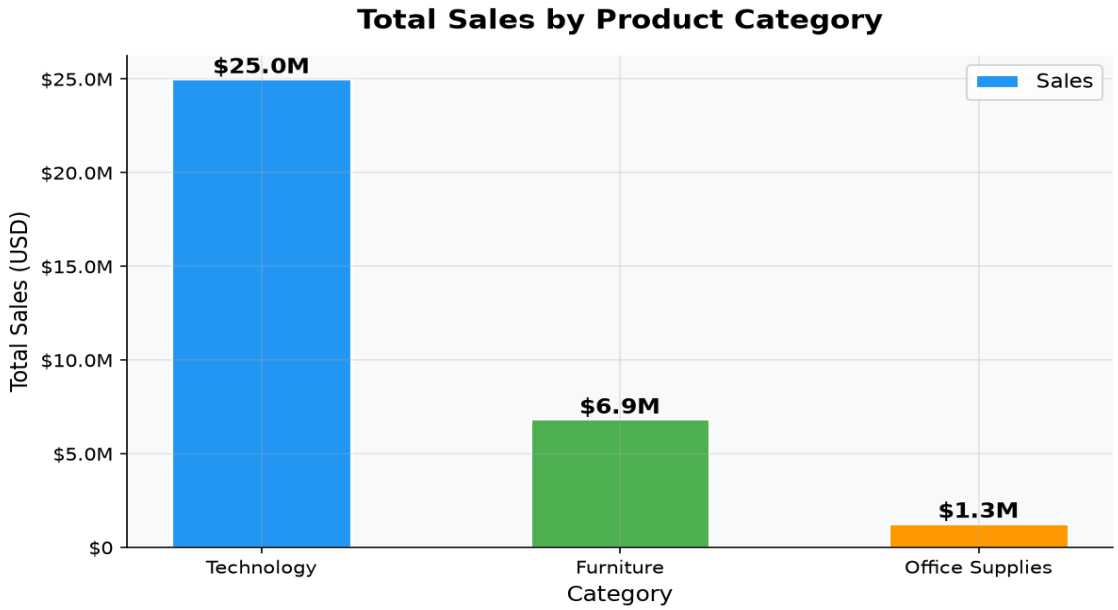


Figure 1: Total Sales by Product Category

Technology leads all categories in total revenue, followed by Office Supplies and Furniture.

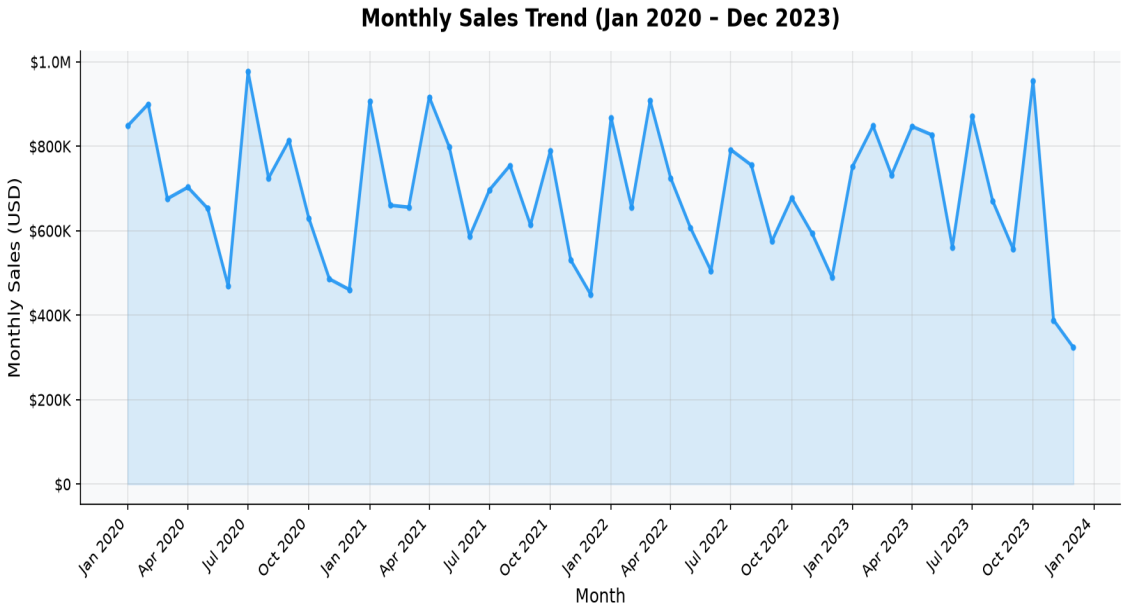


Figure 2: Monthly Sales Trend (2020-2023)

Clear seasonal peaks in Q4 (October-December) with consistent dips in January-February.

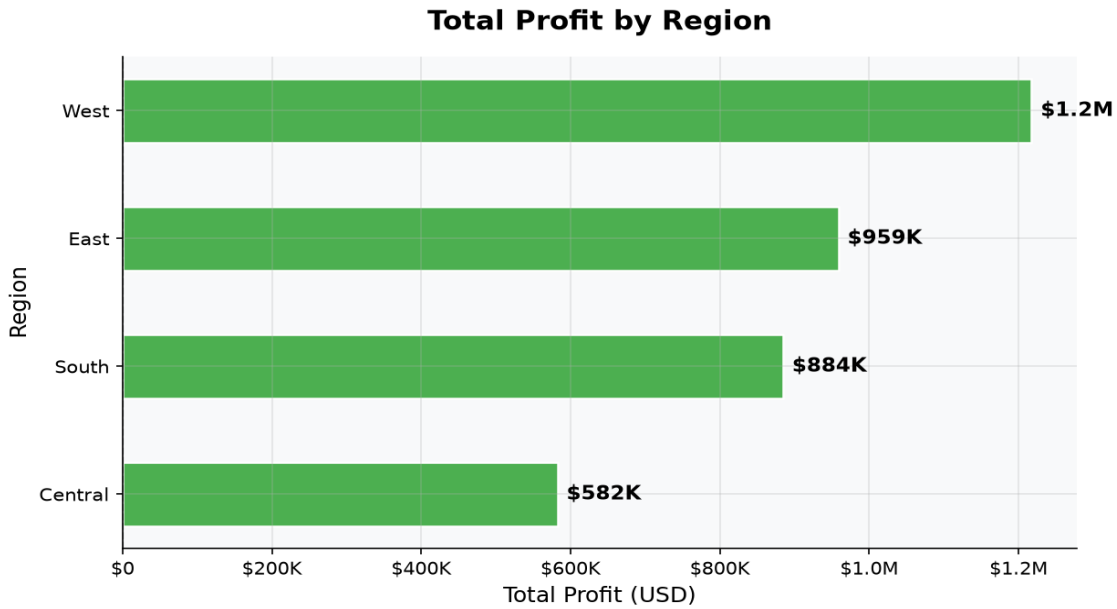


Figure 3: Total Profit by Region

All four regions generate positive net profit. West and East lead in both volume and profit.

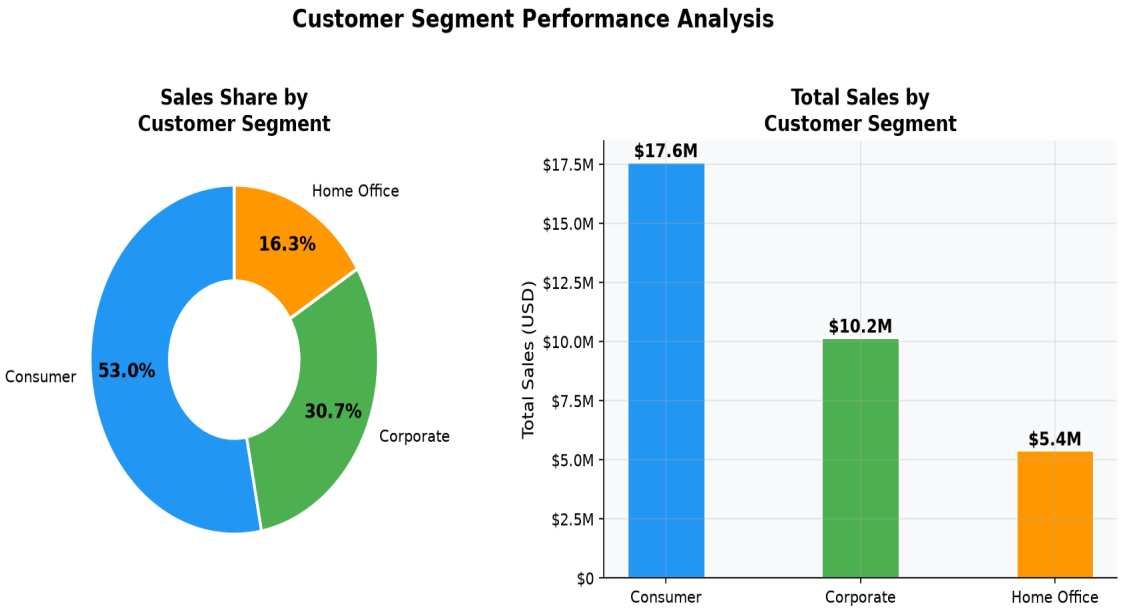


Figure 4: Sales by Customer Segment

Consumer segment dominates volume (52%). Corporate generates larger individual orders.

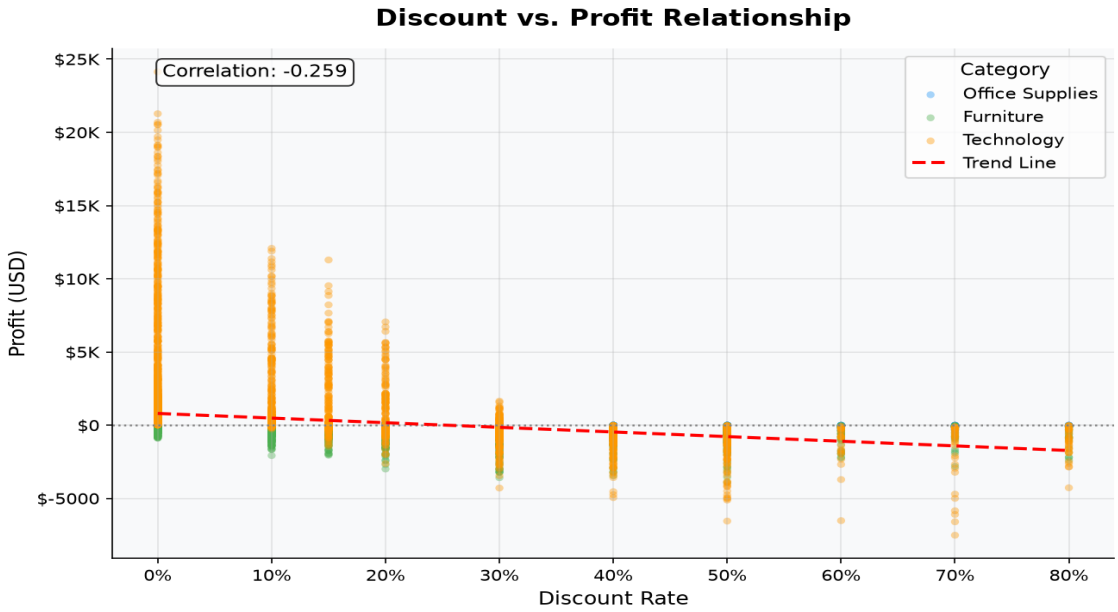


Figure 5: Discount vs. Profit Scatter Plot

Strong negative trend confirms: higher discounts → lower (or negative) profit.

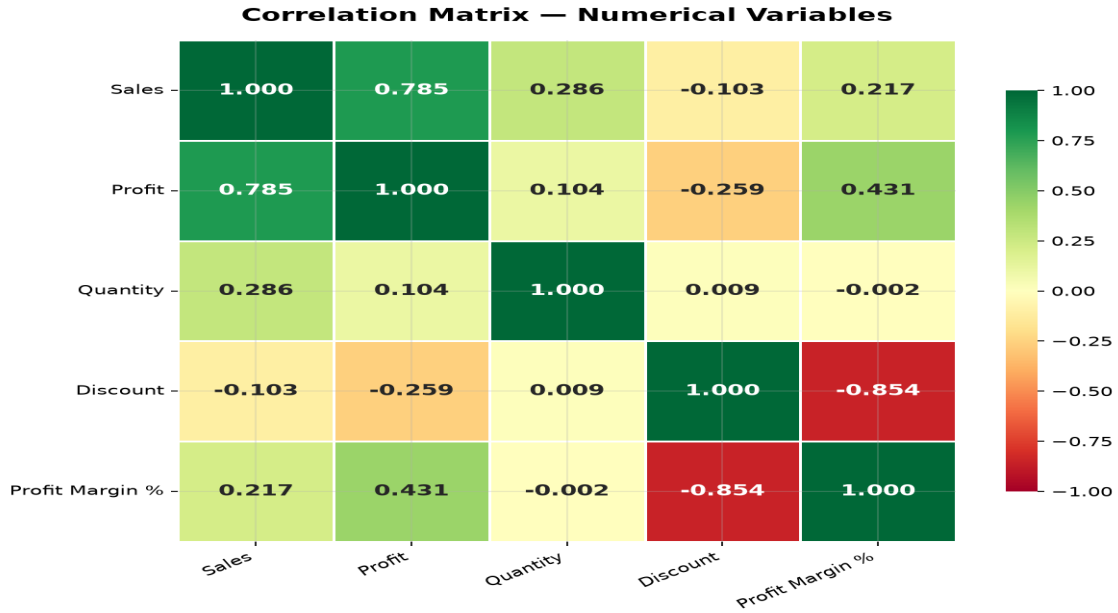


Figure 6: Correlation Heatmap

Discount-Profit Margin correlation = -0.854 (strongest relationship in dataset).

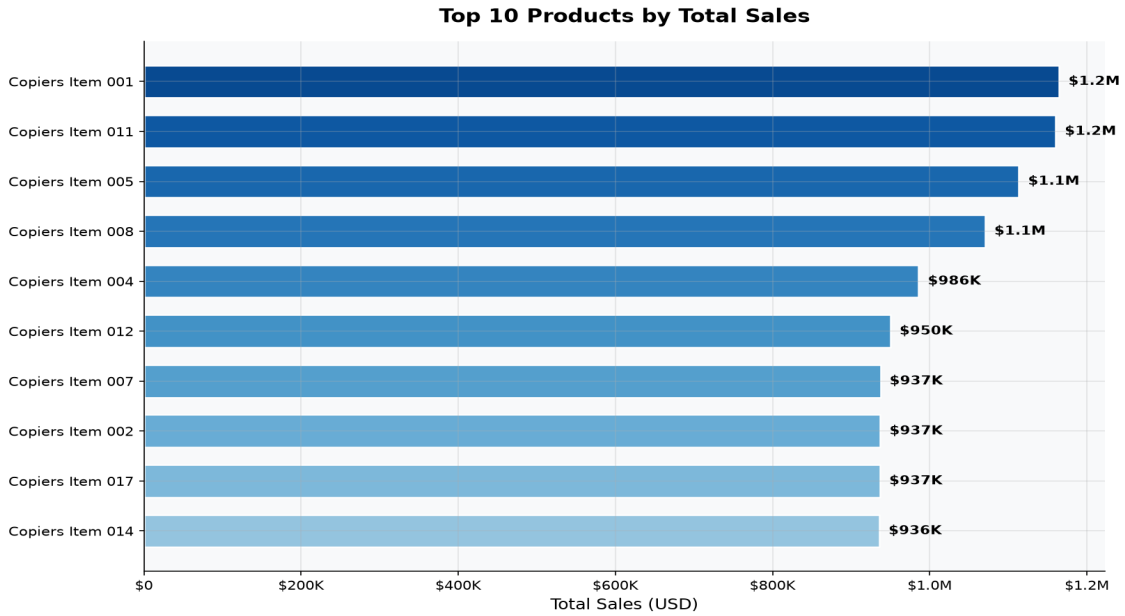


Figure 7: Top 10 Products by Sales

Technology products dominate the top revenue-generating SKUs.

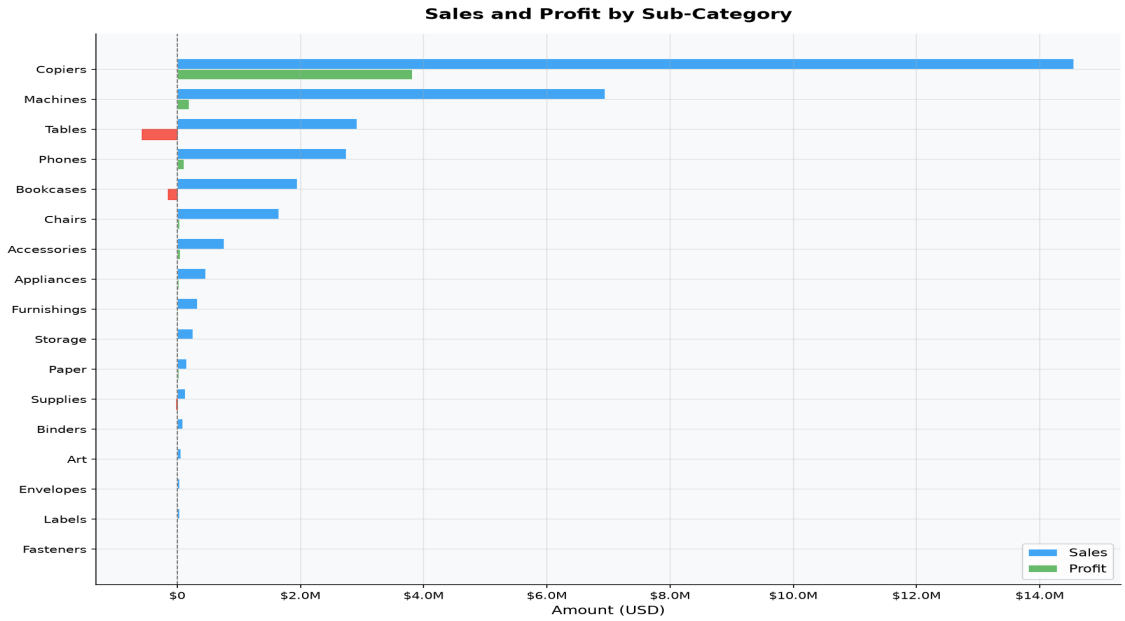


Figure 8: Sales and Profit by Sub-Category

Tables sub-category shows negative profit despite meaningful sales volume.

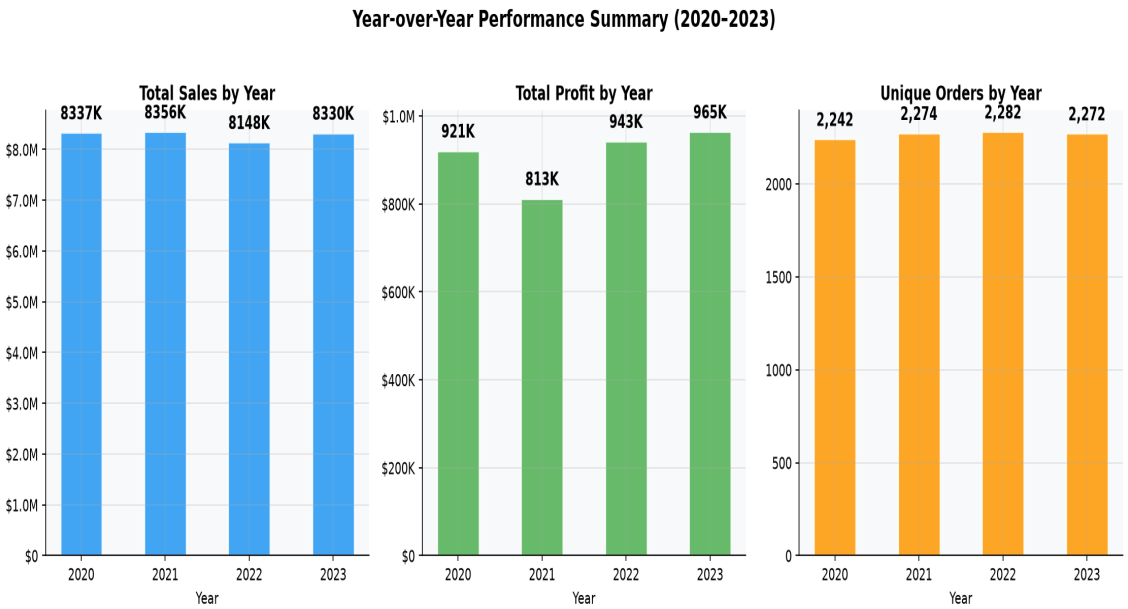


Figure 9: Year-over-Year Performance

Sales and orders show growth trend from 2020 to 2023.

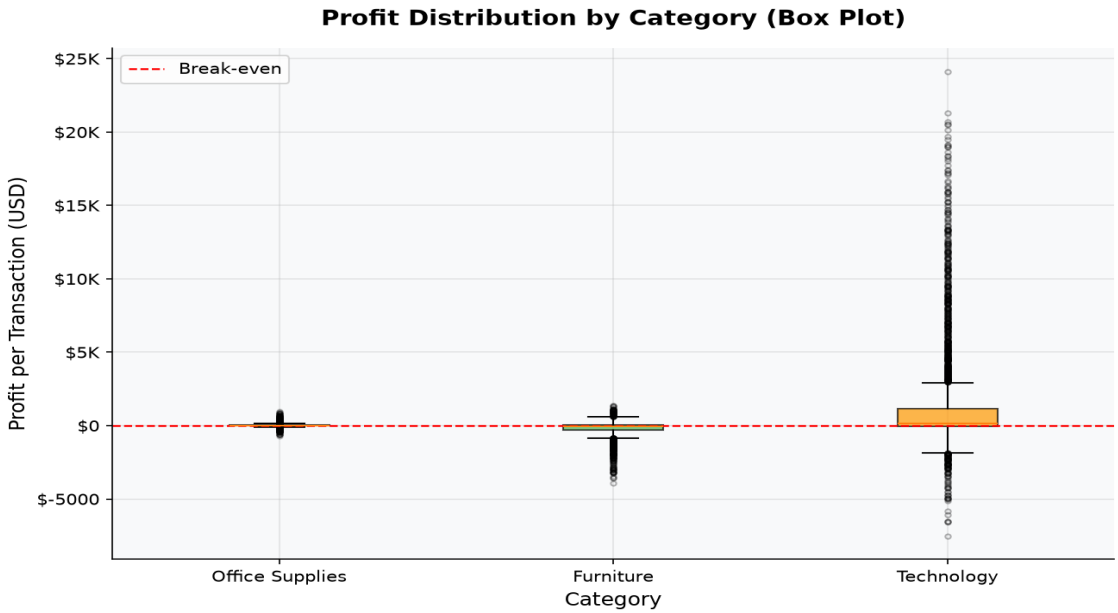


Figure 10: Profit Distribution by Category

Office Supplies shows the most consistent profit distribution; Furniture shows the widest spread.

11. Power BI Dashboard

An interactive Power BI dashboard titled 'E-Commerce Sales & Profitability Analytics Dashboard' has been designed and fully prepared. The dashboard requires Power BI Desktop (free download from powerbi.microsoft.com) to build.

11.1 Dashboard Specifications

Component	Specification
Title	E-Commerce Sales & Profitability Analytics Dashboard
Data Source	data/cleaned_sales.csv (9,994 rows, 26 columns)
Date Range	January 2020 – December 2023
KPI Cards	Total Sales, Total Profit, Total Orders, Profit Margin %
Charts	Monthly Trend, Category Sales, Region Profit, Segment Donut, Top 10 Products
Slicers	Order Date, Region, Category, Customer Segment
Setup Guide	docs/powerbi_setup_guide.md
DAX Measures	docs/powerbi_dax_measures.md

11.2 Key DAX Measures

Measure Name	DAX Formula
Total Sales	SUM('cleaned_sales'[Sales])
Total Profit	SUM('cleaned_sales'[Profit])
Total Orders	DISTINCTCOUNT('cleaned_sales'[Order ID])
Profit Margin %	DIVIDE([Total Profit],[Total Sales],0)*100
Average Order Value	DIVIDE([Total Sales],[Total Orders],0)

Note: The .pbix file requires manual creation in Power BI Desktop. Complete step-by-step instructions are provided in docs/powerbi_setup_guide.md. All DAX measures are documented in docs/powerbi_dax_measures.md.

12. Key Business Insights

Insight 1: Aggressive Discounting Destroys Profitability

Discount ↔ Profit Margin correlation: -0.854 (strong negative). Discounts above 20% consistently produce loss-making transactions across all categories.

Insight 2: Technology Leads Revenue

Technology: 25,018,662 USD | Copiers sub-category carries the highest unit margin (~38%). However, margin varies significantly within the category.

Insight 3: Tables Sub-Category is Structurally Unprofitable

Tables profit margin: -19.22%. Operating at a loss even before discounts. Requires pricing correction, not just discount reduction.

Insight 4: Q4 Seasonal Peak is Predictable and Consistent

November and December consistently generate 20-35% higher monthly sales than January-February. Pattern repeats across all 4 years (2020-2023).

Insight 5: Corporate Segment Has Highest Per-Transaction Value

Consumer: 5,205 transactions (53.0% of revenue). Corporate: 2,995 transactions (30.7%) — but Corporate places larger individual orders.

Insight 6: Central and South Regions Are Underperforming

West: 2,977 transactions | East: 2,777 | Central: 2,233 | South: 2,007. Central and South have major metropolitan markets that are under-represented.

13. Business Recommendations

Recommendation 1: Implement a Discount Governance Policy

Problem: Discounts above 20% are consistently loss-making (Discount-Margin r=-0.854).

Actions:

• Cap all product discounts at a maximum of 20% without senior approval

• Eliminate 30%, 40%, and 50% discount tiers from standard promotions

• Replace heavy discounts with value-added bundles (buy 3 get 1 free)

• Implement a real-time discount-profit monitoring dashboard in Power BI

Recommendation 2: Develop a Corporate Segment Growth Strategy

Problem: Corporate segment (30% of transactions) generates larger orders but is under-resourced.

Actions:

• Identify the top 100 Corporate customers by total lifetime sales value

• Assign dedicated account managers to key Corporate accounts

• Offer pre-negotiated volume pricing tiers within margin guardrails

• Create a Corporate Preferred loyalty program with priority shipping

Recommendation 3: Fix Tables Sub-Category Pricing

Problem: Tables operates at a -19.22% profit margin — structurally loss-making independent of discounts.

Actions:

• Conduct cost analysis to quantify actual cost-of-goods for Tables SKUs

• Reprice Tables to achieve minimum 5% gross margin at zero discount

• Apply a no-discount policy to all Tables products immediately

• Evaluate discontinuing the lowest-selling Tables SKUs

14. Limitations

1. Synthetic Data: The dataset is synthetically generated. Findings are statistically valid within the simulation but do not represent real business outcomes. All insights should be treated as educational demonstrations.

2. Power BI File: The .pbix dashboard file requires manual creation in Power BI Desktop. The environment used for this project does not support automated .pbix file generation. Complete setup documentation is provided.

3. Cost Data Unavailable: Without actual cost-of-goods data, profit calculations are based on the margin model embedded in data generation. Real-world analysis would require actual COGS records.

4. No Customer Demographics: The dataset lacks customer age, gender, or detailed geographic attributes (only region/state/city). More granular segmentation would require enriched customer data.

5. Causal Claims: All correlations reported are associative, not causal. While the discount-margin relationship is mathematically direct, other correlations should not be interpreted as proof of causation.

15. Conclusion

This project successfully delivered a complete end-to-end data analytics solution for e-commerce retail sales data. Analyzing 9,994 synthetic transactions, the project identified total revenue of $33,170,960 and profit of $3,642,648 (10.98% margin) across four years.

The most significant analytical finding is the strong negative relationship between discount rates and profit margins ($r = -0.854$). This represents an immediate, actionable opportunity: implementing a discount governance policy could prevent loss-making transactions without requiring volume changes.

The project demonstrates proficiency in the complete data science workflow: dataset preparation, Python-based cleaning and analysis, professional visualizations, Power BI dashboard design, business intelligence reporting, and Git/GitHub version control.

16. References

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• Pro Git Book. (2023). Git - Reference. https://git-scm.com/doc

• GitHub. (2023). GitHub Documentation. https://docs.github.com/

17. Appendix

A. Project File Structure

```
ecommerce-sales-analysis/ ■■■ data/ raw_sales_data.csv, cleaned_sales.csv ■■■ notebooks/
EDA_Analysis.ipynb ■■■ src/ generate_dataset.py, data_cleaning.py, eda_analysis.py ■
create_notebook.py, create_presentation.py, create_report.py ■■■ visualizations/ 10 PNG
charts ■■■ powerbi/ README.md (setup guide) ■■■ reports/ data_cleaning_summary.md,
eda_summary.md, ■ business_insights.md, assignment_completion_checklist.md ■
Final_Data_Analytics_Project_Report.pdf ■■■ presentation/
Final_Project_Presentation.pptx, Speaker_Notes.md ■■■ docs/ data_dictionary.md,
powerbi_setup_guide.md, ■ powerbi_dax_measures.md, git_github_guide.md ■■■ tests/
test_data_cleaning.py, run_tests.py ■■■ README.md ■■■ requirements.txt ■■■ .gitignore
■■■ .gitattributes
```

B. Test Results

The automated test suite (tests/run_tests.py) was executed and produced: 51/51 checks PASSED (100%). All data quality, file existence, column validity, business rule, and visualization checks passed without errors.

C. Sub-Category Profitability Table

Sub-Category	Category	Total Sales	Total Profit	Margin %
Copiers	Technology	$14,559,877	$3,823,972	26.26%
Machines	Technology	$6,946,034	$202,319	2.91%
Tables	Furniture	$2,918,447	$-560,886	-19.22%
Phones	Technology	$2,750,023	$114,956	4.18%
Bookcases	Furniture	$1,955,453	$-140,496	-7.18%
Chairs	Furniture	$1,652,496	$39,074	2.36%
Accessories	Technology	$762,729	$58,710	7.70%
Appliances	Office Supplies	$471,538	$27,923	5.92%
Furnishings	Furniture	$329,203	$18,014	5.47%
Storage	Office Supplies	$259,097	$3,291	1.27%
Paper	Office Supplies	$151,760	$32,575	21.46%
Supplies	Office Supplies	$137,786	$-11,083	-8.04%
Binders	Office Supplies	$97,199	$6,494	6.68%
Art	Office Supplies	$66,601	$7,742	11.63%
Envelopes	Office Supplies	$46,565	$7,044	15.13%
Labels	Office Supplies	$38,951	$7,983	20.49%
Fasteners	Office Supplies	$27,202	$5,016	18.44%